Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 28)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

0 1 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

0 1 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

0 1 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

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0 2 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 2 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 2 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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0 2 . 3 Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

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Answer [ ] Earths

0 2 . 4 Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 2 . 5 Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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0 3 A student investigated the effect of pH on the germination of cress seeds.

0 3 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

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0 3 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

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0 3 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

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0 3 . 4 The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

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0 4 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 4 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 5 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

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0 5 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

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0 5 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

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0 5 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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0 6 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 6 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 6 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 7 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

0 7 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

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Answer [ ] %

0 7 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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0 7 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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0 7 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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0 8 . 1 Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 8 . 2 Explain how pheromone traps can be used in pest control.

[2 marks]

0 8 . 3 Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

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0 8 . 4 Give **one** way farmers can support crop pollinators.

[1 mark]

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0 9 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 9 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 9 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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1 0 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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1 0 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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1 0 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

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1 0 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

OR

1 1 . 2 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS